ISA 444: Business Forecasting

16 - Autocorrelation and Partial Autocorrelation

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Spring 2022

Outline

1. **Preface**
2. Explaining Sample ACF Plots
3. Partial Autocorrelation
4. Recap

What we Covered Last Class

Main Learning Outcomes

☑ **Define the population mean, and variance of a random variable.**

☑ **Define the population covariance, and correlation between two random variables.**

☑ **Define the population autocovariance and autocorrelation of a random variable.**

☑ **Use sample estimates of the population mean, variance, covariance, and correlation.**

☑ **Explain the properties of the large sample distribution of the sample ACF.**

☑ **Use the large sample distribution of the sample ACF to identify significant autocorrelation in a time series.**

Recap: ACF Definitions

We defined **autocorrelation as the correlation between a time-series and its past values.**

The **ACF was defined as a function that captures the correlations between pairs of values at a certain lag**. For example:

- Lag-1 autocorrelation captures the correlation between y_t and y_{t-1}.
- Lag-2 autocorrelation captures the correlation between y_t and y_{t-2}.

Recap: HW 09 - Autocorrelation

Per your feedback post exam 01, let us discuss the solutions to HW 09 over the next few minutes.

As usual, please do not hesitate to ask me questions.

Recap: The ACF of Advance Retail Sales [RSCCAS]

$\pm 1.96 * \frac{1}{\sqrt{n}}$

Build on the example below and plot the ACF for:

- The RSCCAS series, which is stored in the `retail` df. ts $\Rightarrow \frac{1}{\sqrt{362}}$
- The first differences of the RSCCAS series. diff $\Rightarrow \frac{1}{\sqrt{361}}$

```
pacman::p_load(tidyverse, magrittr, fpp2, tidyquant, scales, ggpubr)
retail = tq_get('RSCCAS', from = ymd('1992-01-01'), to = Sys.Date(),
                get = 'economic.data')
```

Based on both plots, answer the following two questions:

- Are the limits for both ACF plots the same? Why? Why not?
- Comment on how their ACF plots are different.

Learning Objectives for Today's Class

Main Learning Outcomes

- **Determine if a sample ACF plot "cuts off" or "dies down".**
- **Explain how sample partial autocorrelation is calculated.**

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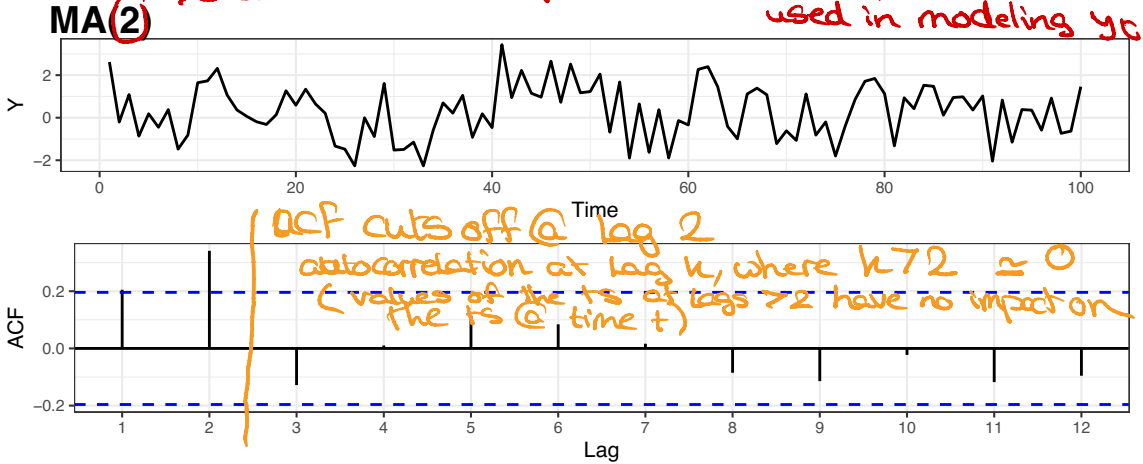
Some Time Series and their ACF Plots [1]

→ order of the MA process ⇒ 2 error terms are used in modeling y_t

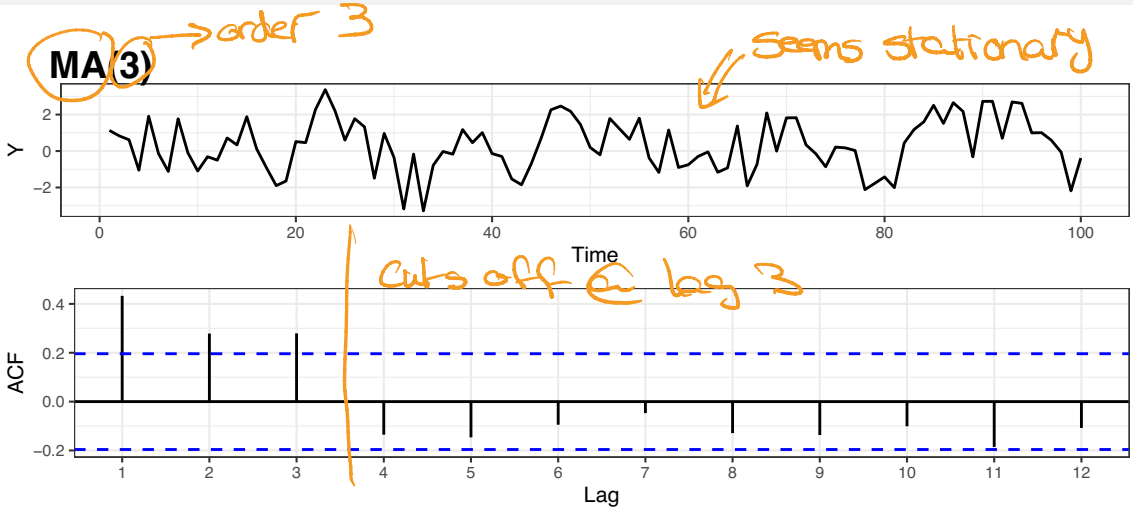


acf cuts off @ lag 2

autocorrelation at lag k, where k > 2 ≃ 0

(values of the ts of lags > 2 have no impact on the ts @ time t)

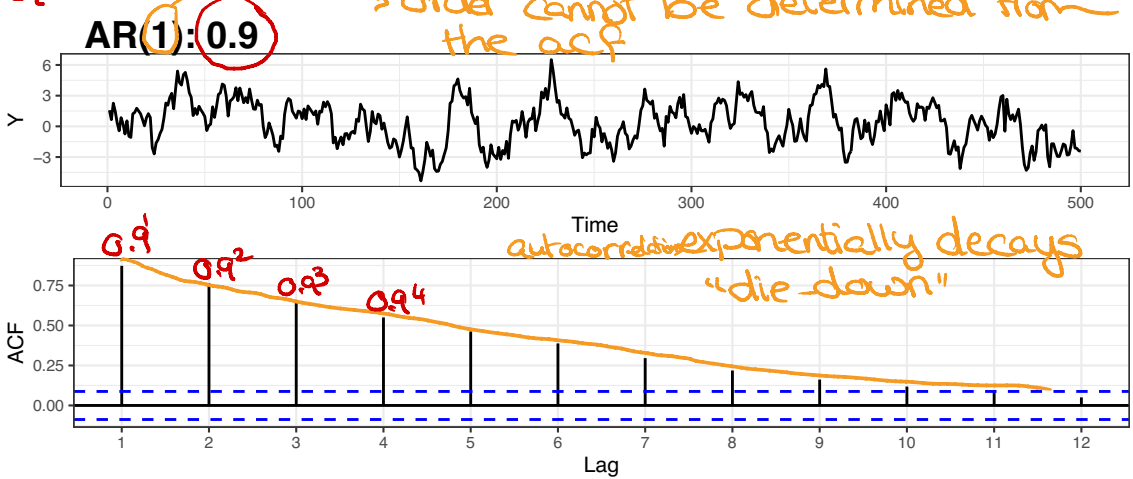
Some Time Series and their ACF Plots [2]



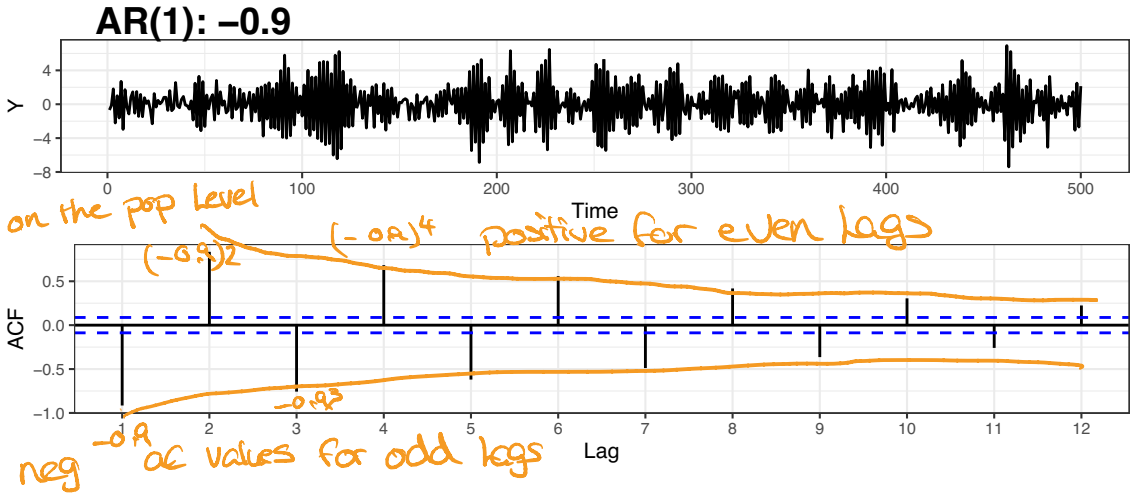
Some Time Series and their ACF Plots [3]

$y_t \sim y_{t-1}$

order cannot be determined from the acf



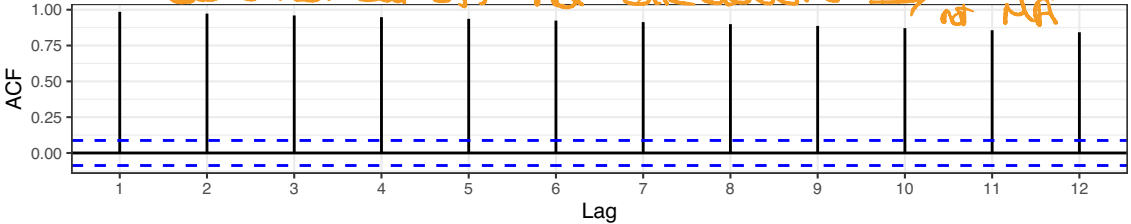
Some Time Series and their ACF Plots [4]



Some Time Series and their ACF Plots [5]

random walk process $y_t = y_{t-1} + \varepsilon$

does not cut off nor die down $\Rightarrow$ not AR not MA



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3 **Partial Autocorrelation**

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Definition: General

Statistical Definition: Let us say that we have three variables, X, Y, and Z, all correlated, and we want to know how X and Y are correlated after we remove the effects of Z on each.

Computation Approach:

$$\hat{X} = a_1 + b_1 Z; \qquad X^* = X - \hat{X}$$

$$\hat{Y} = a_2 + b_2 Z; \qquad Y^* = Y - \hat{Y}$$

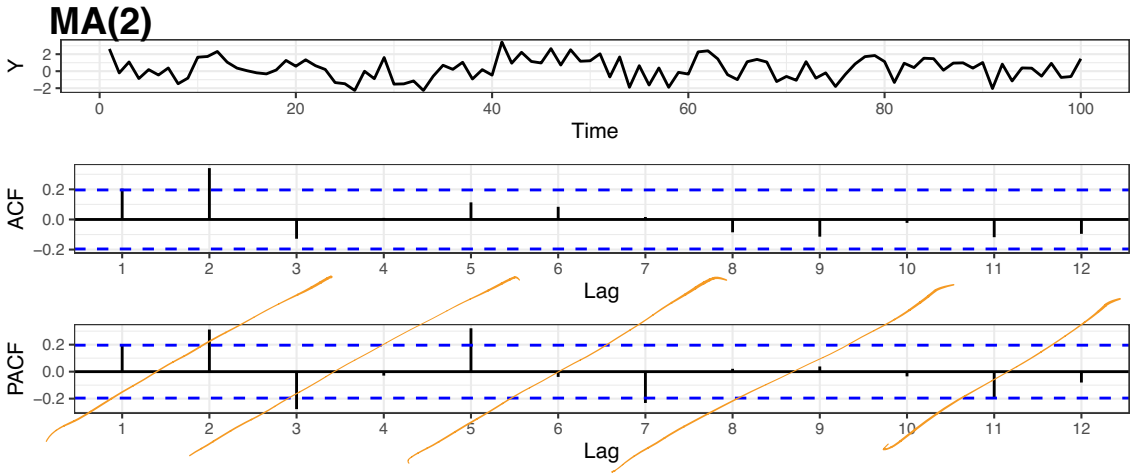
$Corr(X^*, Y^*)$ is the partial correlation between X and Y. It is the correlation that remains after we remove the effect of Z.

PACF in the Context of Time-Series Analysis

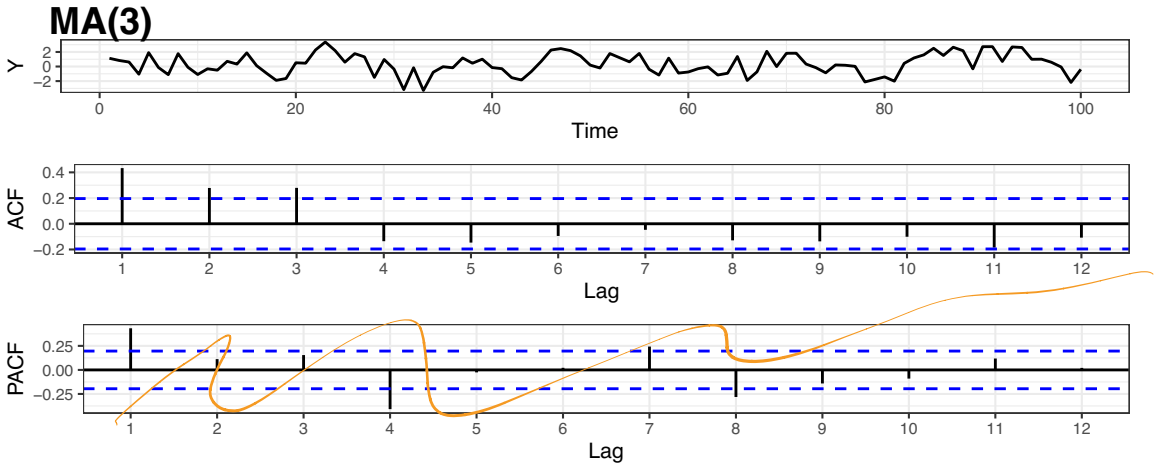
The Partial Autocorrelation between Y_t and Y_{t+k} is the correlation between Y_t and Y_{t+k} after removing the effects of Y_{t+1}, Y_{t+2}, Y_{t+3}, ... , Y_{t+k-1}.

- We plot the partial autocorrelation over multiple lags just like the autocorrelation function (ACF).
- We refer to the plotted partial autocorrelations as the PACF.

Some Time Series and their PACF Plots [1]

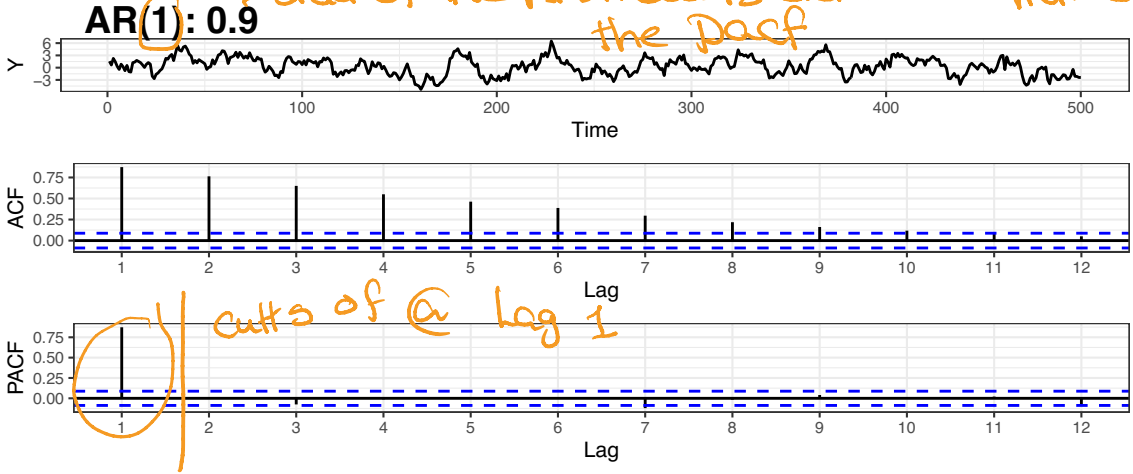


Some Time Series and their PACF Plots [2]



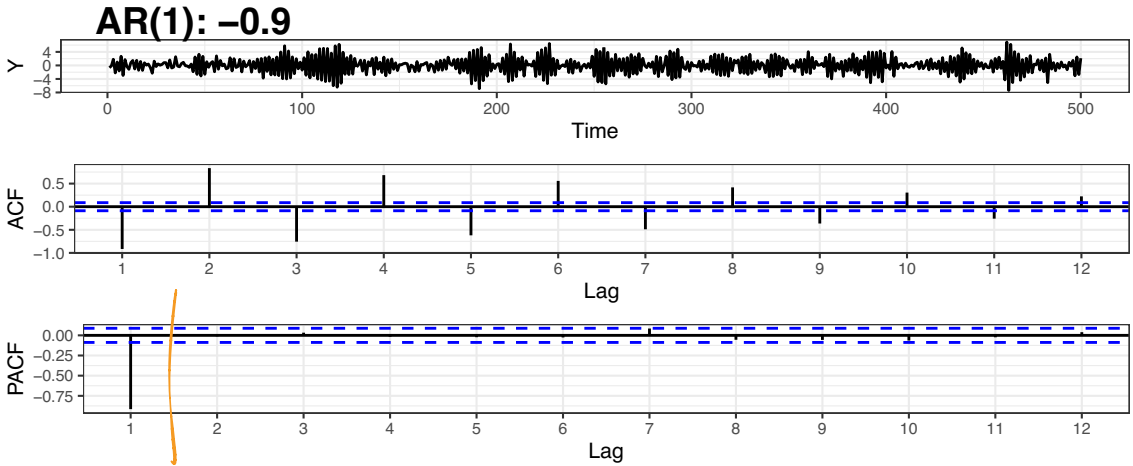
Some Time Series and their PACF Plots [3]

order of the AR model is determined from the PACF



cuts of @ lag 1

Some Time Series and their PACF Plots [4]



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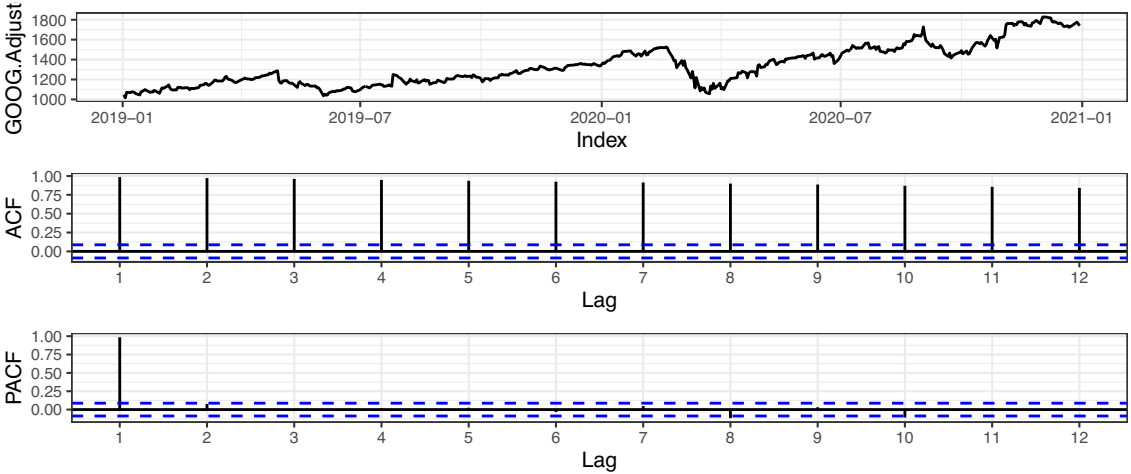
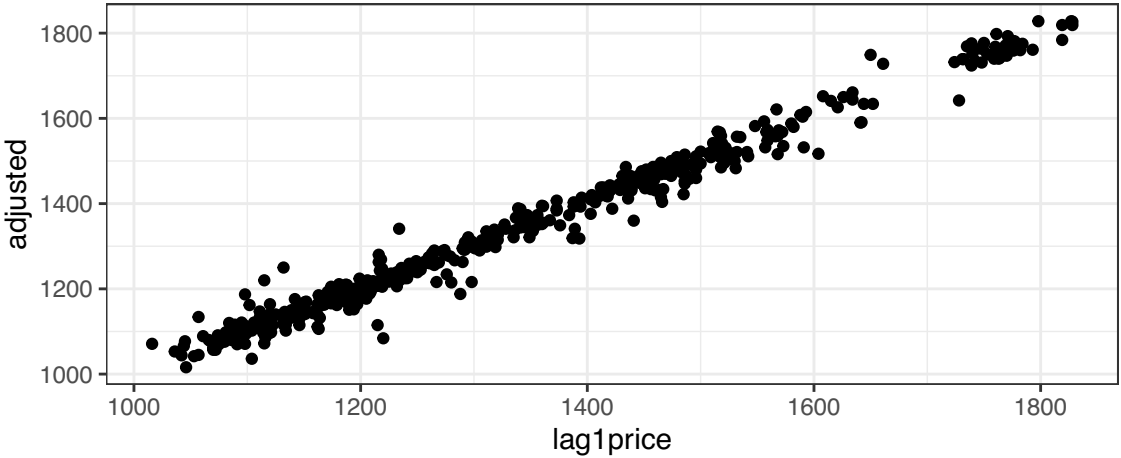


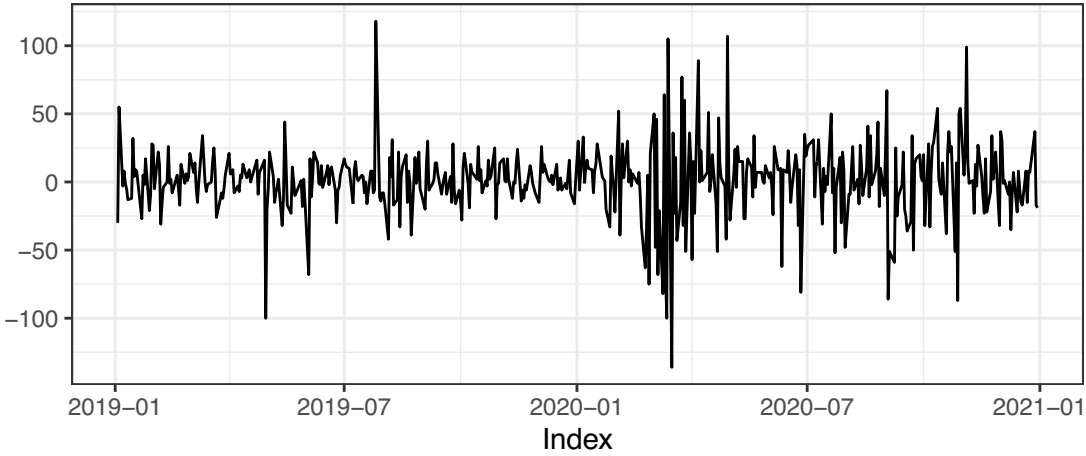
What we have learned from the GOOG Example [1]



What we have learned from the GOOG Example [2]

	Dependent variable:
	adjusted
lag1price	0.993***
	(0.006)
Constant	10.224
	(7.924)
Observations	503
R^2	0.983
Adjusted R^2	0.983
Residual Std. Error	26.271 (df = 501)
F Statistic	28,570.970*** (df = 1; 501)
Note:	*p<0.1; **p<0.05; ***p<0.01

What we have learned from the GOOG Example [3]



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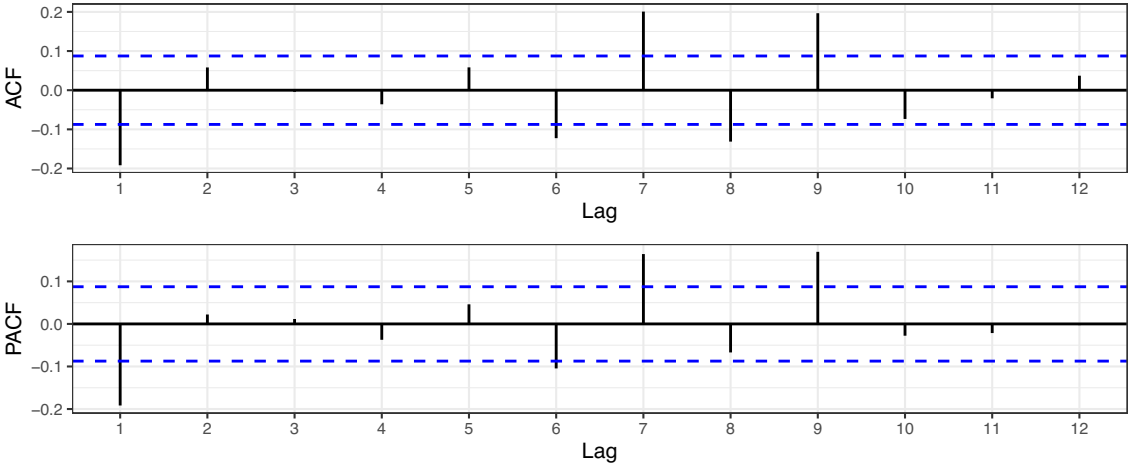

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Summary of Main Points

Main Learning Outcomes

- **Determine if a sample ACF plot "cuts off" or "dies down".**
- **Explain how sample partial autocorrelation is calculated.**

Things to Do

- Go through the slides, examples and make sure you have a good understanding of what we have covered.
- Go through the self-paced tutorial for week 08 (which will take you sometime between 5-10 minutes to complete).
- Complete the assignment (see details in next slide).

Graded Assignment 10: Evaluating your Understanding

Please go to Canvas (click here) and answer the questions. **The assignment is due March 18, 2022 [8:30 AM].**

What/Why/Prep? The purpose of this assignment is to evaluate your understanding and retention of autocorrelation and PACF. To reinforce your understanding of the covered material, I also suggest going through the self-paced tutorial for week 08 (which will take you sometime between 5-10 minutes to complete).

General Guidelines:

- Individual assignment.
- This is **NOT** a timed assignment.
- Proctorio is NOT required for this assignment.
- You will need to have R installed (or accessible through the Remote Desktop)

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